An ODE Compartment Model of Post-College Financial Dynamics

Jay Maroney
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Abstract

Faced with student loans, modest starting salaries, and new expenses, many recent college graduates must decide how to divide income between spending, saving, investing, and debt repayment to build long term financial health. Compartment models described by systems of ordinary differential equations provide a transparent way to track flows between accounts and compare strategies over time. This paper develops a three–compartment ODE model for cash, invested assets, and debt. Using this model, we are able to derive a reduced equation for net wealth and analyze existence, uniqueness, invariance of the non–negative region, and local stability using linearization and eigenvalues. A scalar approximation identifies a threshold parameter that balances income against spending and debt interest and predicts whether wealth converges to a modest plateau or grows rapidly. Numerical simulations for three realistic strategies (low saving with slow repayment, balanced saving and repayment, and aggressive early repayment followed by heavy investing) illustrate how changing spending fractions or repayment intensity qualitatively reshapes trajectories and wealth levels in the long run. The model connects standard personal finance advice to dynamical systems concepts and offers a flexible framework for exploring "what–if" questions about post–college money management.

1 Introduction

Graduating from college typically signifies a transition into the real world and the burdens that come with it, namely financial independence. Many recent graduates face student loans, moderate starting salaries, new expenses such as rent and insurance, and difficult choices about how to balance spending, saving, investing, and debt repayment to maximize long–term financial well–being. With an overload of available resources, choosing a clear personal finance plan can feel overwhelming, which often leads to procrastination or avoidance among young adults.

Compartment models described by systems of ordinary differential equations (ODEs) offer an intuitive framework for representing complex processes in a structured and interpretable way. This project uses a compartment model to track the financial evolution of a recent graduate by dividing wealth into three categories: cash, invested assets, and debt. The goal is to understand how different choices, such as keeping spending high and paying debt slowly versus aggressively repaying loans or investing early, influence whether net wealth grows or stagnates over a multi–decade horizon.

The paper is organized as follows. Section 2 introduces the compartments, parameters, assumptions, and income schedule, as well as derives the governing ODEs together with a reduced equation for net wealth. Section 3 applies existence and uniqueness theory, proves forward invariance of the non–negative region, and analyzes equilibria and local stability using the Jacobian and eigenvalues. Section 4 derives a scalar approximation that highlights a threshold parameter that controls long-term wealth growth. Section 5 presents numerical simulations for three financial strategies and interprets the trajectories and phase portraits in light of the analysis. Section 6 discusses key takeaways for graduates and limitations and extensions of the model.

2 Model Setup and Assumptions

2.1 Compartments and Variables

The model follows a single individual over time t, measured in years since graduation. At each time, three state variables represent the person's financial position:

- $C(t)$: checking or cash balance.
- $A(t)$: invested assets, such as retirement accounts, brokerage accounts, and savings accounts.
- $D(t)$: total outstanding debt, including student loans and credit card balances, assumed non–negative.

Net wealth is defined as

$$W(t) = C(t) + A(t) - D(t).$$

Time–dependent income enters the checking account at a rate $I(t)$, a piecewise constant function that represents salary raises at discrete times.

2.2 Parameters and Financial Assumptions

Flows between compartments are governed by constant fractions over the time horizon:

- a: fraction of cash spent on necessary expenditures (food, utilities, transportation).
- b: fraction of cash spent on rent or housing.
- c: fraction of cash contributed to investments and savings.
- d: fraction of cash devoted to debt repayment.
- e: average annual yield rate of all investment and savings accounts (APY).
- f: annual fraction of assets liquidated back into cash.
- g: average annual interest rate on debt (APR).
- r: typical percentage salary raise.

The outflow fractions from the checking compartment satisfy

$$a + b + c + d = 0.95,$$

so that a small remaining fraction of income serves as an automatic buffer or safety net for shocks and modeling error.

2.3 Key Modeling Assumptions

To keep the model tractable yet realistic, several simplifying assumptions are adopted:

- The individual is continuously employed, so $I(t)$ is piecewise constant.
- The debt APR g is constant and known, and financing terms do not change.
- Taxes are already accounted for in $I(t)$, which represents post–tax income.
- Inflation is ignored by measuring all quantities in fixed (real) dollars.
- Debt cannot become negative; reaching $D(t) = 0$ corresponds to entering a debt–free regime where repayment flows can be redirected.
- Cash cannot become negative; any purchase is treated as being made with credit if necessary, which increases debt rather than allowing $C(t)$ to cross below zero.

These assumptions parallel common simplifications in epidemic and population models, where positivity and conservation constraints play a central role.

2.4 Income Schedule

Income grows as the graduate advances in a career, with raises at prescribed times t_1, t_2, t_3. The effective income function is

$$I(t) = \begin{cases} I_0, & 0 \le t < t_1, \\ I_0(1 + r_1), & t_1 \le t < t_2, \\ I_0(1 + r_1)(1 + r_2), & t_2 \le t < t_3, \\ I_0(1 + r_1)(1 + r_2)(1 + r_3), & t \ge t_3, \end{cases}$$

where I_0 is initial post–tax income and r_1, r_2, r_3 are raise percentages at each promotion time. Figure 1 shows the resulting stepwise income profile over a 25–year horizon.

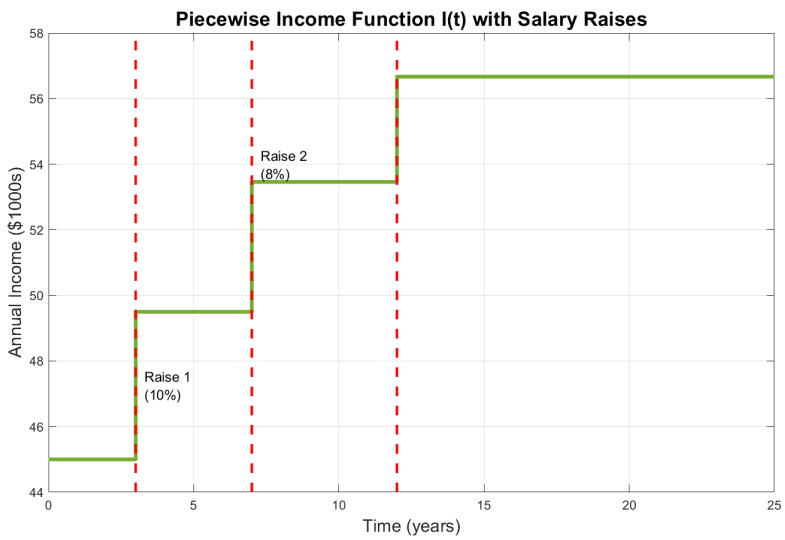

Figure 1: Piecewise income function $I(t)$ with three salary raises over a 25–year horizon.

2.5 Derivation of the Differential Equations

Changes in cash, assets, and debt are determined by the balance of inflows and outflows for each compartment.

2.5.1 Cash $C(t)$

The checking balance increases due to income $I(t)$ and withdrawals from invested assets at rate $fA(t)$, and it decreases through spending, rent, transfers into assets, and debt repayment at total proportional rate $(a + b + c + d)$. Thus,

$$C'(t) = I(t) + fA(t) - (a + b + c + d)C(t).$$

2.5.2 Assets $A(t)$

Assets grow when cash is invested at rate $cC(t)$ and through investment returns at rate $eA(t)$, and they shrink when funds are liquidated at rate $fA(t)$:

$$A'(t) = cC(t) + eA(t) - fA(t) = cC(t) + (e - f)A(t).$$

2.5.3 Debt $D(t)$

Debt grows due to interest at rate $gD(t)$ and decreases when cash is used to repay loans at rate $dC(t)$. This is modeled as

$$D'(t) = gD(t) - dC(t).$$

2.5.4 Net Wealth $W(t)$

Net wealth is

$$W(t) = C(t) + A(t) - D(t),$$

so

$$W'(t) = C'(t) + A'(t) - D'(t).$$

Substituting the expressions above and simplifying yields

$$\begin{aligned} W'(t) &= I(t) + fA(t) - (a+b+c+d)C(t) + cC(t) + eA(t) - fA(t) \\ &\quad - \big(gD(t) - dC(t)\big) \\ &= I(t) - (a+b)C(t) + eA(t) - gD(t). \end{aligned}$$

This compact expression shows that wealth rises when income and directed debt repayment outweigh both the interest on debt and the combined spending and rent fractions.

3 Mathematical Analysis

On each time interval between salary raises, income is constant, so the system is autonomous on that subinterval. Denoting the constant income on a given interval by I_x, the system becomes

$$C' = I_x + fA - (a+b+c+d)C, \tag{1}$$

$$A' = cC + (e-f)A, \tag{2}$$

$$D' = gD - dC. \tag{3}$$

This section analyzes existence and uniqueness of solutions, invariance of the non–negative octant, and local behavior near equilibria.

3.1 Existence, Uniqueness, and Non–Negativity

Write the system as $x' = F(x)$ with $x = (C, A, D)^T$. Each component of F is an affine function of the form $Ax + b$ for a constant matrix A and vector b, so

$$F(x) - F(y) = A(x - y)$$

for all $x, y \in \mathbb{R}^3$. Any matrix norm $\|\cdot\|$ satisfies

$$\|F(x) - F(y)\| \le \|A\| \, \|x - y\|,$$

showing that F is globally Lipschitz and hence satisfies the hypotheses of the existence and uniqueness theorem for systems of ODEs. It follows that for any initial condition $(C(t_0), A(t_0), D(t_0))$

there is a unique solution defined for all $t \geq t_0$ on each income interval, and these solutions patch together to form a unique global trajectory across raises.

Financially meaningful states satisfy

$$C(t) \geq 0, \quad A(t) \geq 0, \quad D(t) \geq 0.$$

The model is constructed so that the non–negative octant is forward invariant:

- If $C = 0$ and $A, D \geq 0$, then from (1),

$$C' = I_x + fA \geq 0,$$

  so cash cannot cross into negative values.

- If $D = 0$ and $C \geq 0$, then from (3),

$$D' = -dC,$$

  so crossing $D = 0$ corresponds to entering a debt–free regime in which repayment flows can be redirected into investments or spending. In this model, repayment is redirected to investments.

- If $A = 0$ and $C, D \geq 0$, then from (2),

$$A' = cC \geq 0,$$

  so invested assets remain non–negative.

- If $A = 0$, $C \geq 0$, and $D = 0$ then,

$$A' = (c + d)C \geq 0.$$

This forward–invariance logic parallels standard arguments for positivity in compartment models from population biology and epidemiology.

3.2 Equilibria for Fixed Income

On an income interval where $I(t) = I_x$ is constant, an equilibrium is a triple (C^*, A^*, D^*) with $C' = A' = D' = 0$. Assuming $D \geq 0$, substituting into (1)–(3) gives

$$\begin{aligned} 0 &= I_x + fA^* - (a + b + c + d)C^*, \\ 0 &= cC^* + (e - f)A^*, \\ 0 &= -dC^* + gD^*. \end{aligned}$$

From the asset equation,

$$A^* = -\frac{c}{e - f}C^* \quad \text{if } e \neq f.$$

To satisfy $A^* \geq 0$ and $C^* \geq 0$ simultaneously, either $c = 0$ or $e - f < 0$; otherwise the algebraic equilibrium lies outside the non–negative region. Since the realistic financial parameters satisfy $e - f < 0$, the algebraic equilibrium obtained from the linear system typically lies outside the non-negative region. This behavior is common for linear systems with constant inputs: algebraic equilibria exist in $\mathbb{R}^3$ but may not lie in the physically meaningful region.

In practice, the model is therefore used to compare trajectories and approximate wealth levels rather than to insist on a physically realizable equilibrium triple (C^*, A^*, D^*).

3.3 Linearization and Jacobian

To understand local behavior near a rest state, we linearize the system via the Jacobian matrix. Writing $x' = F(x)$ with $x = (C, A, D)^T$, we have

$$DF(x) = \begin{pmatrix} -(a+b+c+d) & f & 0 \\ c & e-f & 0 \\ -d & 0 & g \end{pmatrix}.$$

Because F is affine, this Jacobian is constant, so the linearization about any equilibrium x^* is

$$y' = Jy, \quad J = \begin{pmatrix} -(a+b+c+d) & f & 0 \\ c & e-f & 0 \\ -d & 0 & g \end{pmatrix},$$

where $y = x - x^*$. The linearization theorem states that if none of the eigenvalues of J have zero real part, then the phase portrait of the nonlinear system near x^* is qualitatively the same as that of the linear system $y' = Jy$.

3.4 Eigenvalues and Stability Classification

The Jacobian J has a block structure: the cash–asset dynamics appear in the upper–left 2×2 block

$$J_{CA} = \begin{pmatrix} -(a+b+c+d) & f \\ c & e-f \end{pmatrix},$$

while the debt dynamics appear only in the third row and column. One eigenvalue is immediately

$$\lambda_3 = g,$$

associated with the debt direction; with a positive APR, $\lambda_3 > 0$.

The remaining eigenvalues are those of J_{CA}, whose characteristic polynomial is

$$p(\lambda) = \lambda^2 - \tau\lambda + \Delta,$$

where

$$\tau = \mathrm{tr}(J_{CA}) = (e-f) - (a+b+c+d), \quad \Delta = \det(J_{CA}) = -(a+b+c+d)(e-f) - fc.$$

The eigenvalues are

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}.$$

For a planar linear system $y' = J_{CA}y$, the Routh–Hurwitz conditions state that asymptotic stability requires $\tau < 0$ and $\Delta > 0$. In this model,

- $\tau < 0$ means $e - f < a + b + c + d$: effective asset growth is smaller than the total proportional outflow from checking.
- $\Delta > 0$ means $-(a+b+c+d)(e-f) - fc > 0$, which further restricts how large $e - f$ can be while still damping perturbations in the cash–asset subsystem.

When both conditions hold, the cash–asset subsystem is a stable node or spiral, so small deviations in C and A decay back toward equilibrium. The sign of $\tau^2 - 4\Delta$ determines whether trajectories approach monotonically or with oscillations.

For the full 3×3 system the eigenvalues are $\lambda_1, \lambda_2, \lambda_3$. Because $\lambda_3 = g > 0$, any global equilibrium is a saddle point: nearby solutions may be attracted along the stable C–A directions (if $\lambda_{1,2} < 0$) but are repelled in the debt direction. The stable manifold corresponds qualitatively to policies that keep D small, whereas the unstable directions capture the tendency of uncontrolled debt to grow under positive interest.

4 Reduced Wealth Equation and Scalar Threshold

Net wealth is defined as

$$W(t) = C(t) + A(t) - D(t),$$

so differentiating yields

$$W'(t) = C'(t) + A'(t) - D'(t).$$

Substituting from the governing equations gives

$$\begin{aligned} W'(t) = &\big[I(t) + fA(t) - (a+b+c+d)C(t)\big] \\ &+ \big[cC(t) + (e-f)A(t)\big] \\ &- \big[gD(t) - dC(t)\big]. \end{aligned}$$

Collecting like terms, we obtain the exact reduced wealth equation

$$\boxed{W'(t) = I(t) - (a+b)C(t) + eA(t) - gD(t).}$$

This expression shows that net wealth increases through income and investment returns, while it decreases through spending, rent, and debt interest. The asset term $eA(t)$ contributes positively to wealth growth, whereas the debt term $gD(t)$ acts as a persistent drag.

Scalar Approximation

On an income interval where $I(t) = I_x$ is constant, numerical simulations indicate that the relative proportions of cash, assets, and debt tend to stabilize over time before debt is fully extinguished. We therefore approximate

$$C(t) \approx \alpha W(t), \quad A(t) \approx \beta W(t), \quad D(t) \approx \delta W(t),$$

where $\alpha, \beta, \delta \geq 0$ represent typical proportions satisfying approximately

$$\alpha + \beta - \delta \approx 1.$$

In later debt–free regimes, $\delta \approx 0$ and the approximation effectively reduces to a two–compartment cash–asset balance.

Substituting these expressions into the reduced wealth equation yields

$$\begin{aligned} W'(t) &\approx I_x - (a+b)\alpha W(t) + e\beta W(t) - g\delta W(t) \\ &= I_x - \Lambda W(t), \end{aligned}$$

where the effective drag coefficient is

$$\boxed{\Lambda := (a+b)\alpha - e\beta + g\delta.}$$

The resulting scalar equation

$$W'(t) = I_x - \Lambda W(t)$$

has the explicit solution

$$W(t) = \left(W(t_0) - \frac{I_x}{\Lambda} \right) e^{-\Lambda(t-t_0)} + \frac{I_x}{\Lambda}, \quad \Lambda \neq 0.$$

If $\Lambda > 0$, wealth converges to the finite plateau

$$W_\infty \approx \frac{I_x}{\Lambda}.$$

If Λ is small and positive, wealth grows almost linearly over realistic time horizons. If $\Lambda < 0$, the exponential term grows and the reduced model predicts sustained, accelerating wealth growth.

Connection to Eigenvalue Analysis

The positive contribution of assets in the reduced wealth equation is directly related to the eigenvalues of the cash–asset subsystem analyzed in Section 3. In particular, the effective asset growth rate $e - f$ appears in the trace

$$\tau = (e - f) - (a + b + c + d)$$

of the Jacobian block J_{CA}. When $e - f$ is large relative to the total proportional outflow from cash, the dominant eigenvalue of J_{CA} moves toward zero or becomes positive, indicating that asset accumulation increasingly drives system behavior. In this regime, the term $eA(t)$ reduces the effective threshold Λ, enabling sustained or accelerating wealth growth. Conversely, when $e - f$ is small, the cash–asset subsystem is strongly stable, the influence of $A(t)$ is damped, and wealth converges toward a finite level.

5 Numerical Experiments

To explore how different money–management strategies play out over time, the model is simulated over a 25–year horizon for three distinct scenarios:

$$C(0) = \$2{,}000, \quad A(0) = \$0, \quad D(0) = \$30{,}000.$$

The income function $I(t)$ and raise schedule are held fixed across strategies, while the spending, investing, and repayment fractions differ. For each scenario, trajectories of $C(t)$, $A(t)$, $D(t)$, and $W(t)$ are computed and compared with the scalar threshold analysis.

5.1 Scenario 1: Low Saving, Slow Debt Repayment

In the first scenario the graduate maintains relatively high spending and rent and dedicates only small fractions to investment and repayment:

$$a = 0.50, \quad b = 0.25, \quad c = 0.10, \quad d = 0.10, \quad a + b + c + d = 0.95,$$

with asset return $e = 0.04$, liquidation fraction $f = 0.02$ (so $e - f = 0.02$), and debt APR $g = 0.06$.

Figure 2 shows the trajectories of cash, assets, debt, and net wealth for this strategy.

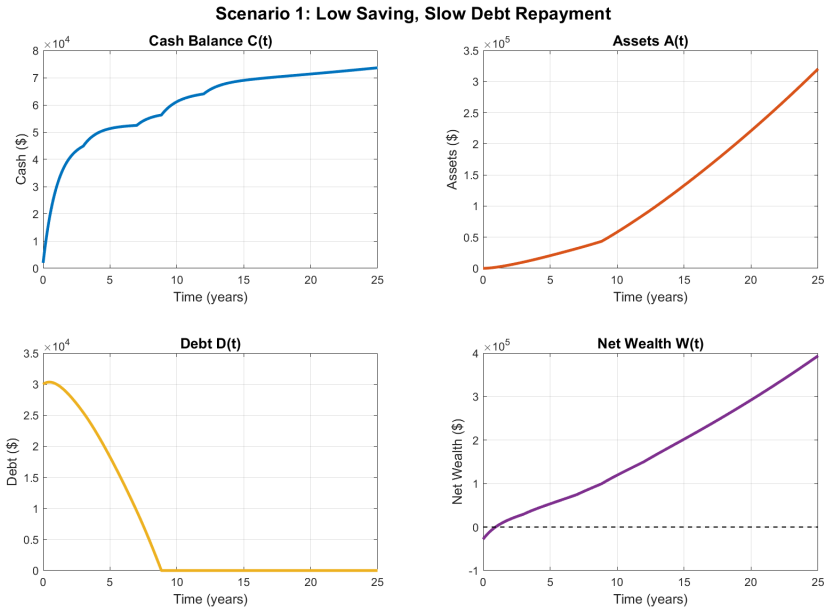

Figure 2: Scenario 1: low saving and slow debt repayment.

Qualitatively, debt declines gradually and is not eliminated until roughly year nine, significantly later than in the other strategies. Assets initially grow slowly but accelerate once debt is eliminated and more cash can flow into investments. Cash initially rises but eventually stabilizes around 70,000 dollars as inflows and outflows approach balance. Net wealth gradually rises in an approximately linear fashion throughout the time horizon.

5.2 Scenario 2: Balanced Saving and Repayment

In the second strategy the graduate deliberately trims spending and rent and allocates more income toward both investing and repayment:

$$a = 0.35, \quad b = 0.20, \quad c = 0.20, \quad d = 0.20.$$

Debt APR remains $g = 0.06$. The stronger investment focus raises the average APY to $e = 0.06$ while the liquidation fraction drops to $f = 0.01$, so $e - f = 0.05$.

Figure 3 displays the resulting trajectories.

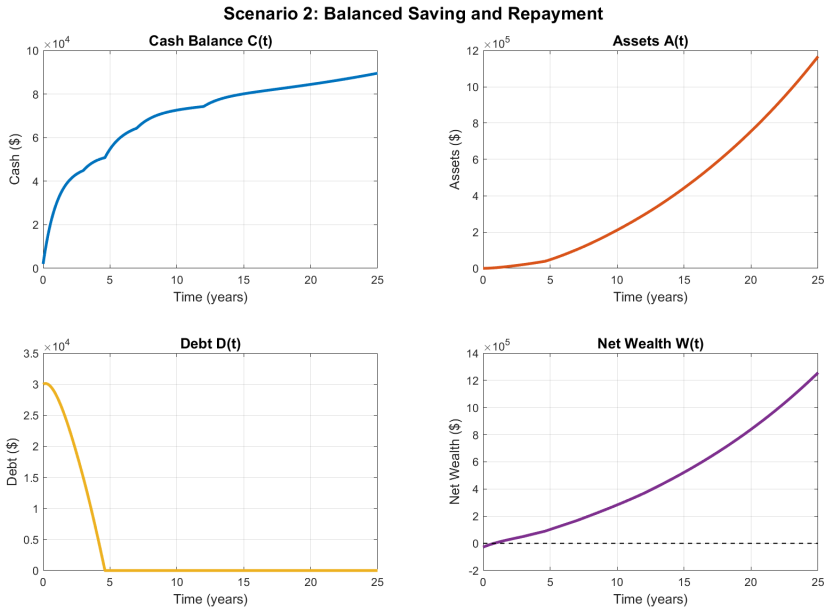

Figure 3: Scenario 2: balanced saving and repayment.

Debt now declines at a sharper rate and reaches zero a bit before 5 years, as the higher repayment fraction d strongly counteracts the gD term in (3). Assets grow significantly due to both higher contributions cC and a better effective return $e - f$, yielding much faster compounding growth compared to the previous scenario. Cash still fluctuates but at a higher average level as debt obligations shrink, and net wealth follows a shallower exponential path due to $gD(t)$ becoming negligible and investing dominating.

5.3 Scenario 3: Aggressive Early Repayment then Investing

The third strategy features two phases: aggressive early repayment followed by heavy investing once debt is low. In Phase 1 (roughly the first five years),

$$a = 0.30, \quad b = 0.20, \quad c = 0.05, \quad d = 0.40,$$

so a large fraction of cash goes toward loans. In Phase 2 (after $D(t)$ falls below a threshold or after $t = 5$ years),

$$a = 0.35, \quad b = 0.20, \quad c = 0.30, \quad d = 0.10.$$

Debt APR remains $g = 0.06$, while the increased investing raises the APY to $e = 0.07$ with the same liquidation fraction f, giving $e - f \approx 0.06$.

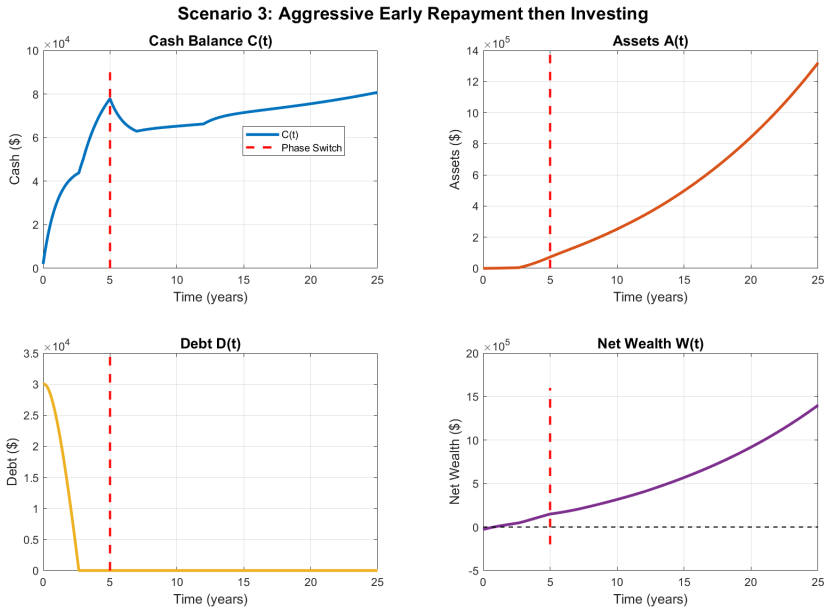

Figure 4: Scenario 3: aggressive early repayment followed by heavy investing. The dashed line marks the phase switch from repayment–focused to investment–focused behavior.

In Phase 1, debt decreases rapidly and falls to zero in around two and a half years since the repayment coefficient d makes it so that D' is dominated by a strong linear decay term in addition to the interest term gD. Assets grow only slightly because c is small, and net wealth is roughly flat or slowly increasing as most free cash goes toward debt. After the switch, assets now receive a large contribution cC and, combined with a relatively high return e, grow rapidly, while net wealth enters a steeper compounding phase.

5.4 Comparative Results Across Scenarios

To compare strategies directly, Figure 5 plots all state variables for each scenario on common axes.

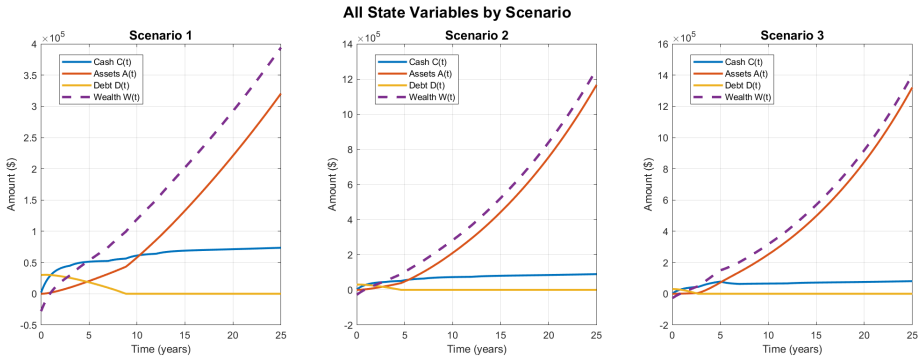

Figure 5: All state variables by scenario: trajectories of cash, assets, debt, and net wealth over 25 years for low saving, balanced, and aggressive–repayment strategies.

Figures 6 and 7 show the debt and asset trajectories across all scenarios, while Figure 8 compares net wealth directly.

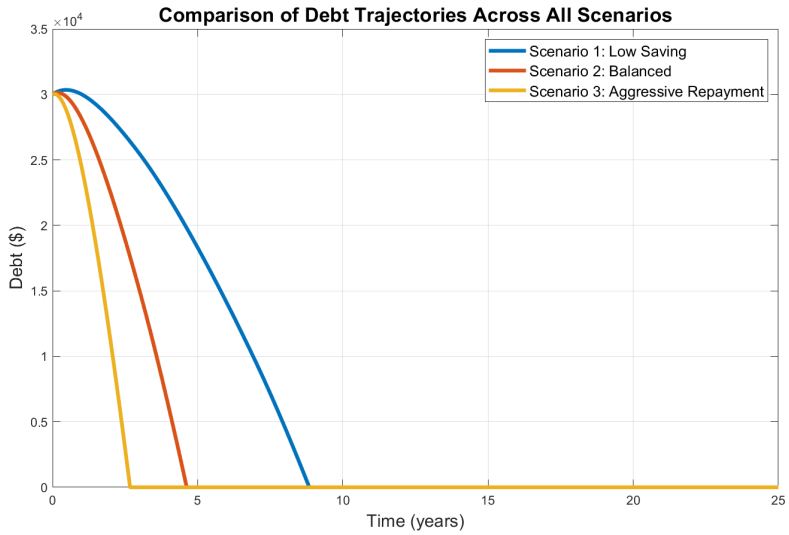

Figure 6: Debt trajectories across all scenarios. Aggressive early repayment eliminates debt fastest, while low saving and low repayment keep debt for nearly a decade.

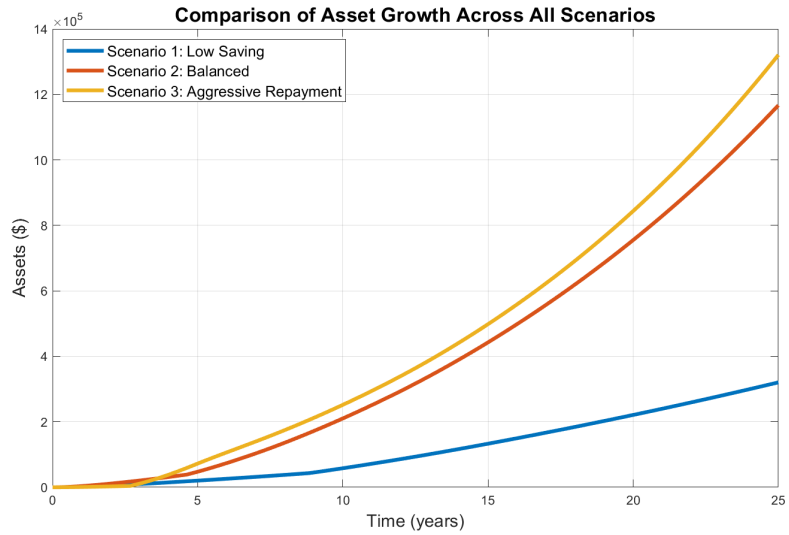

Figure 7: Asset growth across scenarios. Early and sustained investing in the balanced and aggressive strategies leads to much higher asset levels than the low–saving strategy.

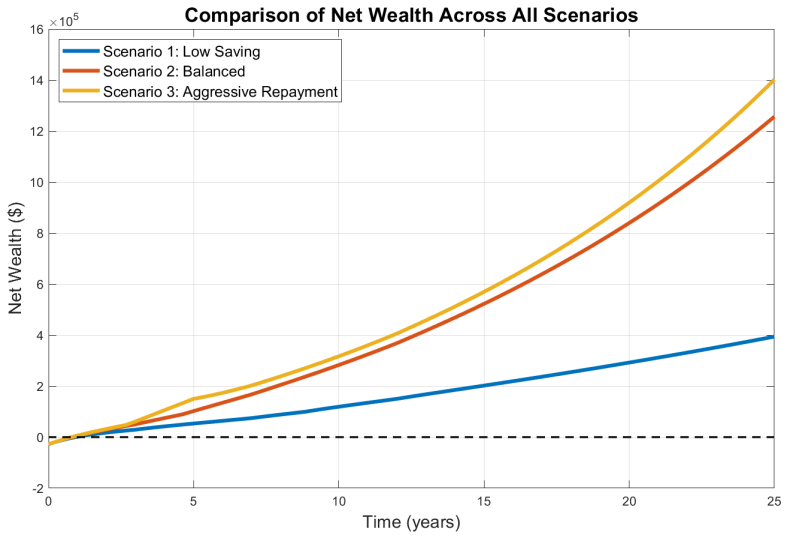

Figure 8: Net wealth trajectories across scenarios. The aggressive early–repayment strategy ultimately achieves the highest net wealth, followed closely by the balanced strategy, while the low–saving strategy lags far behind.

Finally, Figure 9 displays phase portraits of debt versus cash, showing how each policy steers the trajectory through the C–D plane.

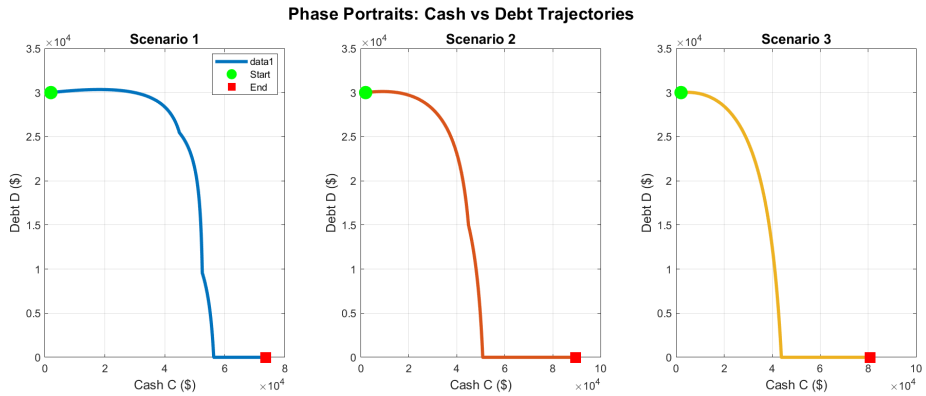

Figure 9: Phase portraits of cash versus debt. Each panel shows the trajectory for one scenario, with the green point indicating the starting state and the red square marking the final state.

Across all comparisons, the low–saving strategy produces persistent debt and relatively low wealth, effectively trapping the system near an undesirable region of the saddle–type dynamics. The balanced strategy reliably moves into a region of low debt and growing assets, yielding a much higher net wealth. The aggressive early–repayment strategy changes the qualitative behavior in the debt direction, enabling earlier arrival at a low–debt manifold and then leveraging heavy investing to accelerate wealth growth.

6 Discussion and Model Improvements

The mathematical analysis shows that on each income interval the financial system is a linear affine ODE with a globally Lipschitz right-hand side, so solutions exist uniquely for all time and remain in the non-negative region for meaningful parameter choices. Linearization and eigenvalues reveal that for realistic positive debt interest the full equilibrium is a saddle: the cash–asset subsystem can be stable, but the debt direction is intrinsically unstable because the associated eigenvalue is $\lambda_3 = g > 0$, independent of the repayment fraction d. Repayment influences trajectories globally through the nonlinear forcing term $-dC(t)$, but it does not change the local linear stability of the equilibrium. The reduced wealth equation

$$W'(t) = I(t) - (a+b)C(t) + eA(t) - gD(t)$$

and its scalar approximation

$$W'(t) \approx I_x - \Lambda W(t)$$

identify Λ as a threshold parameter balancing income against spending, investment growth, and debt interest, where

$$\Lambda = (a+b)\alpha - e\beta + g\delta.$$

Necessary spending and housing costs increase wealth drag, while investment return reduces it and debt interest increases it. Lowering a, b, or g, or increasing the asset fraction β, directly lowers Λ and raises long–run wealth.

The numerical experiments translate this analysis into practical takeaways for recent graduates:

1. **Minimize high–interest debt.** When repayment is weak, the debt direction in the linearized system is unstable, so small increases in debt tend to grow; mathematically, aggressive repayment alters trajectories globally by reducing the forcing term dC(t), pushing the system toward low-debt regions even though the equilibrium remains locally unstable.

2. **Reduce discretionary spending and rent.** The terms $a + b$ increase wealth drag through spending and housing costs, while the investment return e offsets this through asset growth.

3. **Prioritize either repayment or heavy investing, not idle cash.** The successful strategies either maximize early debt repayment or commit strongly to investing once debt is under control; holding large cash balances, as in the low–saving scenario, leads to stagnation or debt traps rather than sustained growth.

At the same time, the model trades realism for ease of use. Income, spending, and returns are treated as smooth flows, whereas in real life they are discrete and stochastic, with biweekly paychecks, occasional big expenses, bonuses, and market volatility. Parameters such as a, b, c, d are taken as fixed fractions, but in reality they evolve with life events such as moving, promotions, or refinancing loans. A natural extension would be to allow parameters to depend on state variables or

time and to perform bifurcation analysis to study how qualitative behavior changes as repayment intensity or investment return vary. Incorporating stochastic shocks or variable interest rates would also improve realism, at the cost of more complicated analysis.

Overall, this project demonstrates how core ideas from this past semester can be used to study real-world financial decisions. The resulting compartment model provides a mathematically grounded yet flexible tool for exploring personal finance strategies and for quantifying the long term consequences of different choices about spending, saving, and debt repayment.

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